

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

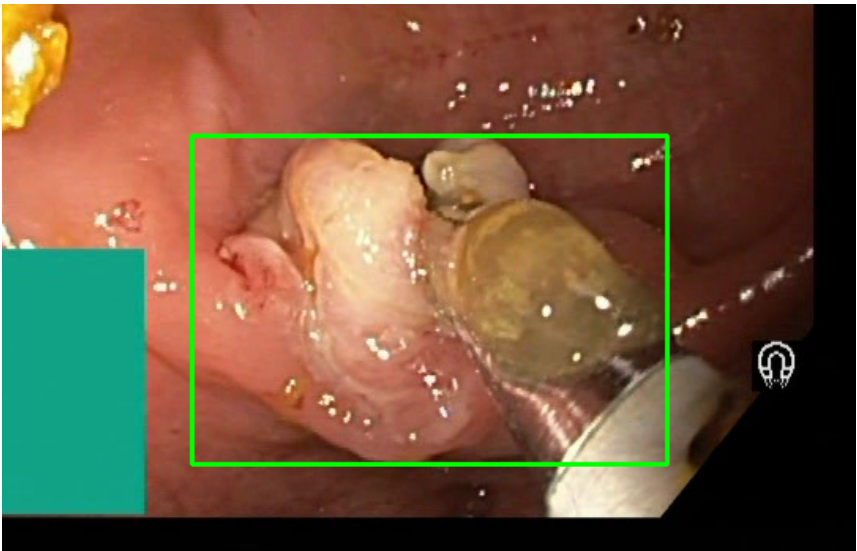
Video ID:	def7538b-640e-4910-bcee-bff38517113b
Total Frames:	5505
Frame Rate:	25.0 fps (assumed)
Duration:	220.2 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	1956	✓
RT-DETR Detections	2533	✓
MedSAM2 Detections	3121	✓
Consensus (3-Model Agreement)	4	✓
Risk Classification	Count	Percentage
HIGH RISK	1	25.0%
MEDIUM RISK	3	75.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	4	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 46 frames)

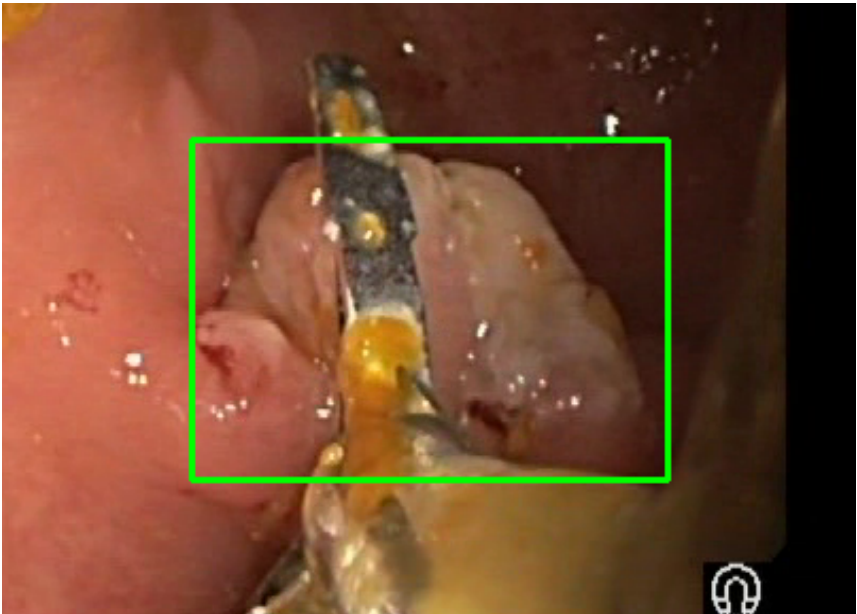


Feature	Value	Interpretation
Frame Index	3058	Frame 3058 of 5505
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	FLAT_POLYP	Type confidence: 88.3%
Redness Score	0.149	Color-based vascularization
Relative Radius	21.7%	Polyp size as % of frame diagonal
Texture Score	0.298	Surface morphology
Vessel Visibility	0.001	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	82.7%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

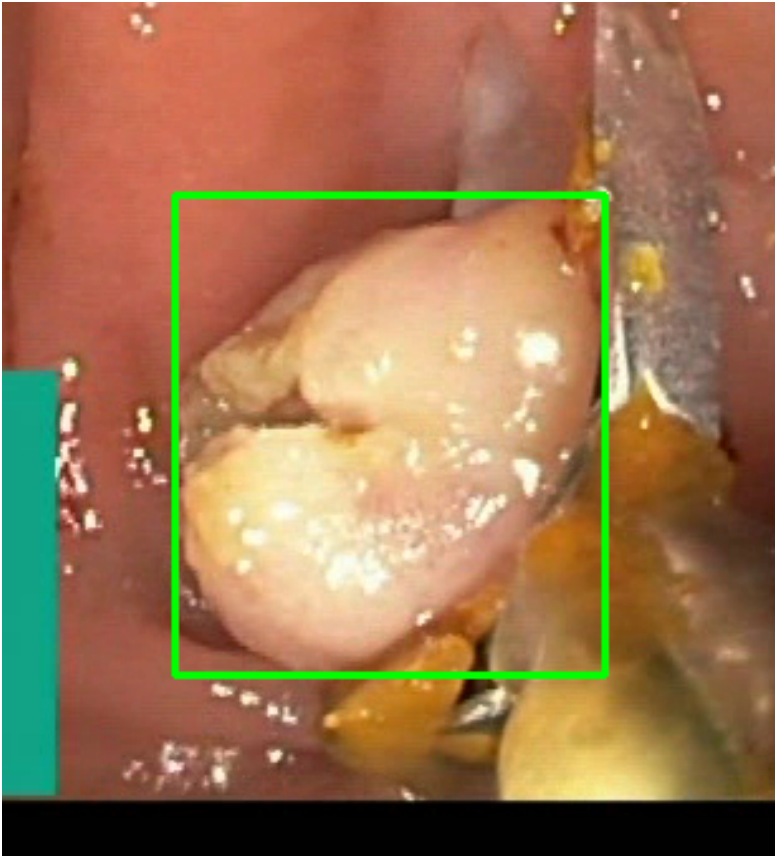
POLYP #2 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 183 frames)



Feature	Value	Interpretation
Frame Index	3415	Frame 3415 of 5505
Detection Method	Detector Consensus (MedSAM2 + RT-DETR)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	FLAT_POLYP	Type confidence: 88.0%
Redness Score	0.154	Color-based vascularization
Relative Radius	19.8%	Polyp size as % of frame diagonal
Texture Score	0.249	Surface morphology
Vessel Visibility	0.001	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	83.8%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

POLYP #3 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 402 frames)



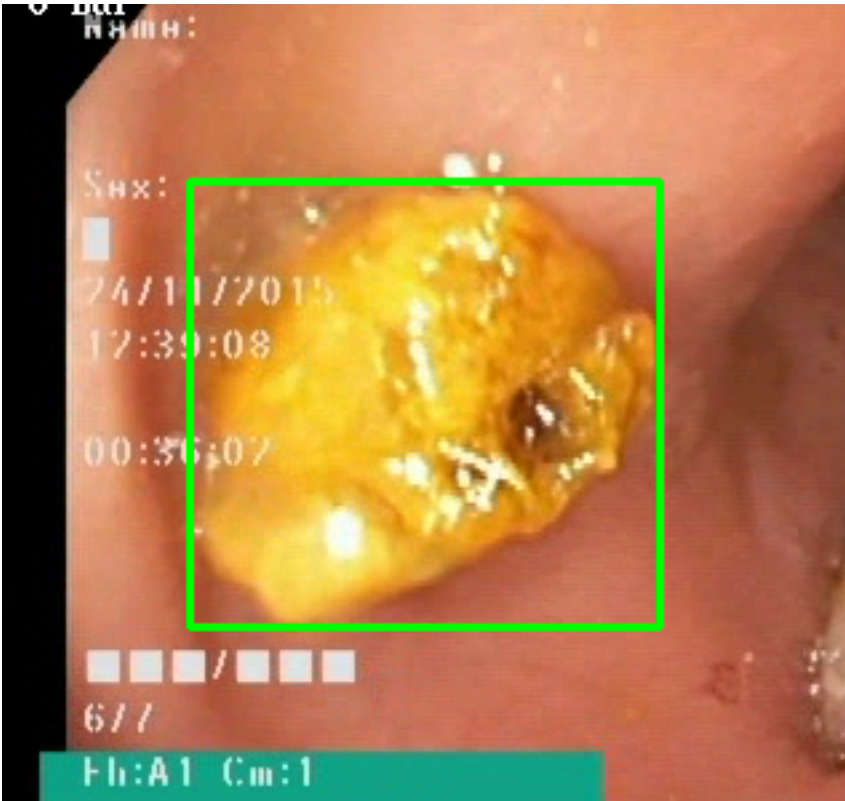
Feature	Value	Interpretation
Frame Index	848	Frame 848 of 5505
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	FLAT_POLYP	Type confidence: 79.0%
Redness Score	0.143	Color-based vascularization
Relative Radius	23.7%	Polyp size as % of frame diagonal
Texture Score	0.150	Surface morphology
Vessel Visibility	0.058	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification

Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	82.6%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #4 — MALIGNANT_POLYP [HIGH_RISK] (Tracked across 110 frames)



Feature	Value	Interpretation
Frame Index	3654	Frame 3654 of 5505
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	MALIGNANT_POLYP	Type confidence: 92.0%
Redness Score	0.178	Color-based vascularization
Relative Radius	20.7%	Polyp size as % of frame diagonal
Texture Score	0.359	Surface morphology
Vessel Visibility	0.004	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output

Clinical Class	MALIGNANT_P OLYP	High suspicion for malignancy — biopsy and urgent resection recommended
Detection Confidence	86.9%	YOLO / RT-DETR / track confidence

Type Assessment: Morphology consistent with high-grade neoplasia — biopsy and urgent resection strongly recommended.

Low-confidence HIGH RISK detection - Further evaluation recommended

CLINICAL IMPRESSION

Summary:

A total of 4 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 1 (25.0%)
- MEDIUM RISK polyps: 3 (75.0%)
- LOW RISK polyps: 0 (0.0%)

Recommendation:

The presence of HIGH RISK polyps warrants close clinical attention. Consider endoscopic resection or follow-up surveillance depending on clinical context.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.